Suwisa Aitha

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Summary

Electrical Engineering M.S. student specializing in Power Systems and Smart Grids, with hands-on experience in MATLAB/Simulink power system dynamics, power quality mitigation (UPQC), and load flow analysis. Proven ability to analyze renewable energy integration, frequency stability, and system performance under disturbances. Experienced in AutoCAD, PowerWorld, Revit, and IoT-based monitoring systems. FE/EIT in progress and open to relocation.

Education

George Mason University — **Fairfax, VA**
M.S. in Electrical Engineering — Jan 2025 – Expected May 2026
-Relevant Coursework: Power System Design, Power System Protection, Power Electronics — GPA: 3.07/4.00

Khon Kaen University — **Khon Kaen, Thailand**
B.S. in Electrical Engineering — Aug 2018 – Jul 2022
-Relevant Coursework: Electrical System Design, Electric Power Systems, Electrical Engineering Laboratory I-III

Experience

Electrical Design and Visualization — **Bangkok, Thailand**
B.S.E. Electronics Co., Ltd. — Cooperative Education Intern — *Jul 2021 – Nov 2021*
-Designed and tested ESP32 microcontroller circuits for real-time electrical data monitoring.
-Programmed Arduino and Unity to synchronize physical and virtual 3D system models, improving visualization efficiency by ~30% over 2D layouts.
-Created 3D electrical layout prototypes that improved coordination with design teams.
-Achieved ~87% alignment between physical and digital models, confirming system accuracy and reliability.

Selected Research Projects

Unified Power Quality Conditioner (UPQC) Simulation Project — **George Mason University**
-Developed a MATLAB/Simulink UPQC model to mitigate voltage sag and harmonic distortion, achieving approximately 50% reduction in THD under disturbed operating conditions.

Power System Dynamics with Renewable Energy Penetration — **George Mason University**
-Modeled a single-machine power system in MATLAB/Simulink and analyzed 0%, 30%, and 60% renewable penetration via inertia reduction, showing deeper frequency nadirs, higher RoCoF, longer settling times, and larger rotor angle oscillations following a three-phase fault.

IoT-Based Air Quality Monitoring System — **George Mason University**
-Reviewed IoT-based embedded systems for PM2.5 and gas detection in Chiang Mai, Thailand, identifying peak PM2.5 levels of 187.6 $\mu g/m^3$ (over 12× WHO limits).
-Proposed a low-cost, scalable, and energy-efficient sensor concept that reduced estimated cost by ≈25% and extended battery life to support community health and sustainability.

Certifications

-FE Exam in Progress — Target EIT Certification by May 2026

Technical Skills

-Programming & Tools: Arduino, Python, Blender (3D Modeling), Microsoft Office, AutoCAD, Revit, PowerWorld Simulator, MATLAB/Simulink
-Professional Skills: Communication, Documentation, Teamwork, Multitasking